
RESEARCH STATEMENT

My research interests center on the semantics of programming languages, computational effects, and categorical semantics. I am interested in how programs and programming constructs acquire formal meaning, how effects are modeled and composed, and how categorical structures can clarify programming-language foundations. Before moving into research, I specialized in air-gapped software at Google Cloud.

EDUCATION

Ashoka University*Bachelor of Science (Honours) in Computer Science*

Major CGPA: 3.74/4

Sonipat, Haryana

Aug 2018 - Apr 2022

RELEVANT COURSEWORK

Discrete Mathematics; Algorithm Design and Analysis; Linear Algebra; Calculus; Abstract Algebra; Probability Theory; Programming Language Design and Implementation; Operating Systems; Computer Networks; Computer Security and Privacy.

EMPLOYMENT

Ashoka University*Research Associate · Programming Languages*

- Studying semantics of programming languages, computational effects, and categorical approaches to programming-language foundations.

Sonipat, Haryana

Jun 2026 - Present

Firebolt*Software Engineer · Self-hosted Platform*

- Joined the founding engineering team building Firebolt's first self-hosted platform for isolated, air-gapped enterprise deployments, including feature flags and Temporal workflow orchestration.

Bangalore, India

Nov 2025 - Mar 2026

Google*Software Engineer · Google Cloud, Distributed Cloud Team*

- Built software for Google Distributed Cloud, bringing Google Cloud services into air-gapped, sovereign, and regulated environments through distributed systems that operated without public-network dependencies.
- Designed a Harbor-based artifact registry for a 40kg portable data center, adding Kubernetes self-healing, node-local fallback mirrors, and image signature verification to support offline bootstrap and operation.
- Built APIs and observability tooling for Prometheus/Grafana monitoring, covering system health, storage, image-pull latency, error rates, alerting, and dashboards for internal teams and deployment operators.
- Mentored junior engineers and hosted a summer intern who later joined full time.

Bangalore, India

Mar 2024 - Sept 2025

Google*Strategic Cloud Engineer · Google Cloud*

- Helped enterprises become cloud-native on Google Cloud, combining technical consulting, hands-on implementation, and architecture guidance across cloud infrastructure projects.
- Researched and prototyped Data Insights, an LLM-powered natural-language layer for tabular data discovery, testing generated questions, SQL guardrails, accuracy, and latency for cold-start exploration of unfamiliar datasets; the capability later became part of a Google Cloud data management product.
- Built reusable cloud and data tooling, including Terraform modules for Kubernetes clusters and VPCs alongside open-source Dataproc templates for Spark/Hadoop ETL.

Bangalore, India

Jul 2022 - Mar 2024

TEACHING

AshokaX*Teaching Assistant · Machine Learning*

- Assisted [Prof. Debayan Gupta](#) in teaching machine learning to high school students.

Online

Apr 2022 - Jun 2022

Ashoka University*Teaching Assistant · Computer Organization and Systems*

- Assisted [Prof. Manu Awasthi](#) with Computer Organization and Systems in Ashoka's Computer Science department.

Sonipat, Haryana

Aug 2021 - Dec 2021

ACTIVITIES

- Participant, [Indian School on Logic and its Applications \(ISLA 2026\)](#), Krea University · Jun 25–30, 2026. Coursework: Programs and Proofs; Computability Theory.
- Participant, [Formal Methods Update Meeting 2026](#), Krea University · Jul 2–3, 2026.